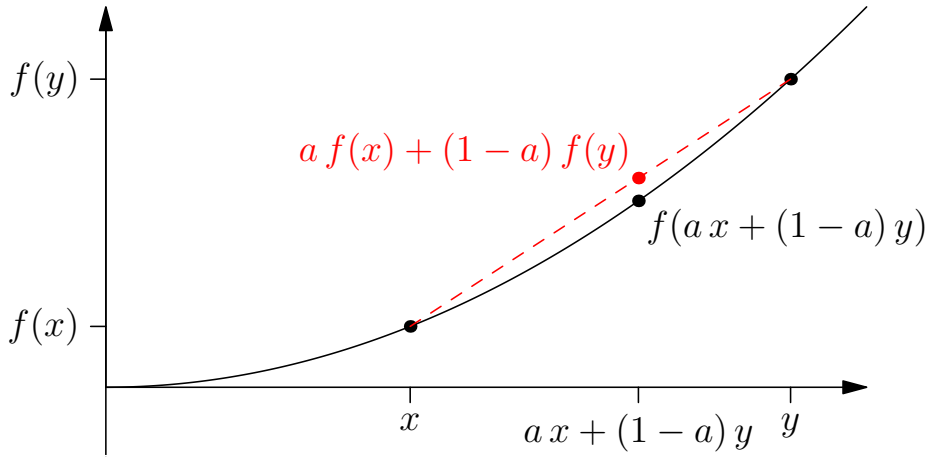
Mathematics of Machine Learning

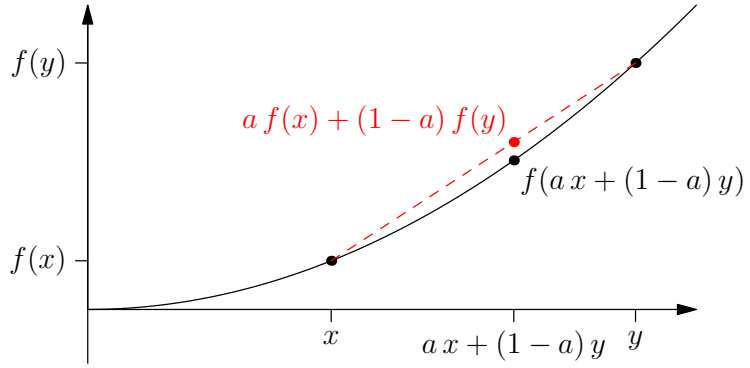
Convexity



Convex sets, convex functions, Jensen's inequality

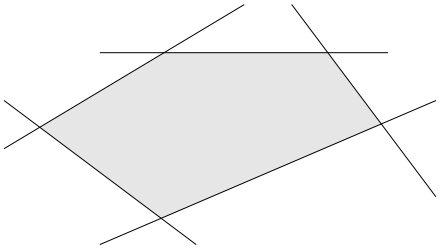
Outline

1. **Convex sets**
2. Convex functions
3. Jensen's inequality



Convex Regions

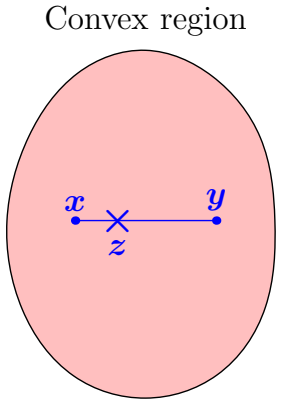
- Convex regions are familiar

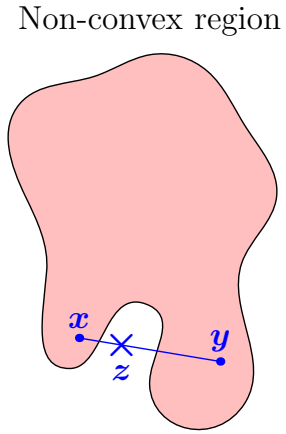


- For any two points $\boldsymbol{x}$ and $\boldsymbol{y}$ in a region $\mathcal{R}$ then for any $a \in [0,1]$ if

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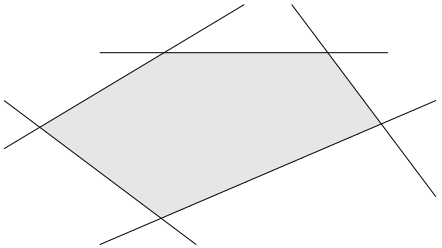
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Convex Regions

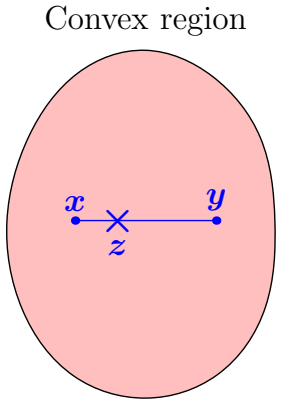
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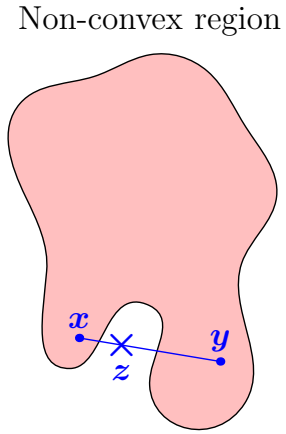


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Convex Sets

- For any set, $\mathcal{S}$, where addition and scalar multiplication are defined (e.g. a vector space) then:

  If for any two elements $\boldsymbol{x},\boldsymbol{y} \in \mathcal{S}$ and any $a \in [0,1]$

  $$\boldsymbol{z} = a\boldsymbol{x} + (1-a)\boldsymbol{y} \in \mathcal{S}$$

  then $\mathcal{S}$ is said to be a convex set

Positive Semi-Definite Matrices

- Recall that a matrix $\mathbf{M}$ is positive semi-definite if for any vector $\boldsymbol{v}$

$$\boldsymbol{v}^{\mathsf{T}}\mathbf{M}\boldsymbol{v} \geq 0$$

- (We showed this also implies that all the eigenvalues are non-negative)
- We denote the fact that $\mathbf{M}$ is positive semi-definite by $\mathbf{M} \succeq 0$, and $\mathbf{M} \succ 0$ if it is positive definite
- The set of positive semi-definite (PSD) matrices (or kernels) form a convex set

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Proof

- Consider any two arbitrarily chosen PSD matrices $\mathbf{M}_1$ and $\mathbf{M}_2$ and any $a \in [0,1]$ then let

$$\mathbf{M}_3 = a\mathbf{M}_1 + (1-a)\mathbf{M}_2$$

- Then for any vector $\boldsymbol{v}$

$$\begin{aligned}\boldsymbol{v}^\mathsf{T}\mathbf{M}_3\boldsymbol{v} &= \boldsymbol{v}^\mathsf{T}(a\mathbf{M}_1 + (1-a)\mathbf{M}_2)\boldsymbol{v} \\ &= a\boldsymbol{v}^\mathsf{T}\mathbf{M}_1\boldsymbol{v} + (1-a)\boldsymbol{v}^\mathsf{T}\mathbf{M}_2\boldsymbol{v} \\ &= am_1 + (1-a)m_2\end{aligned}$$

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- But $m_1, m_2 \geq 0$ since $\mathbf{M}_1, \mathbf{M}_2 \succeq 0$. Thus $am_1 + (1-a)m_2 \geq 0$ and so $\mathbf{M}_3 \succeq 0$ $\square$

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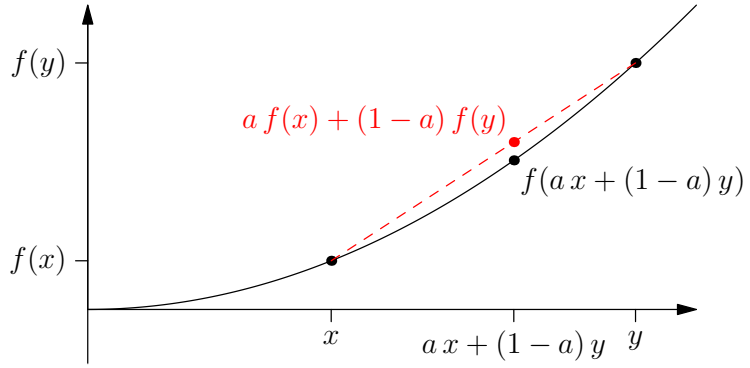
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Outline

1. Convex sets
2. **Convex functions**
3. Jensen's inequality



Convex Functions

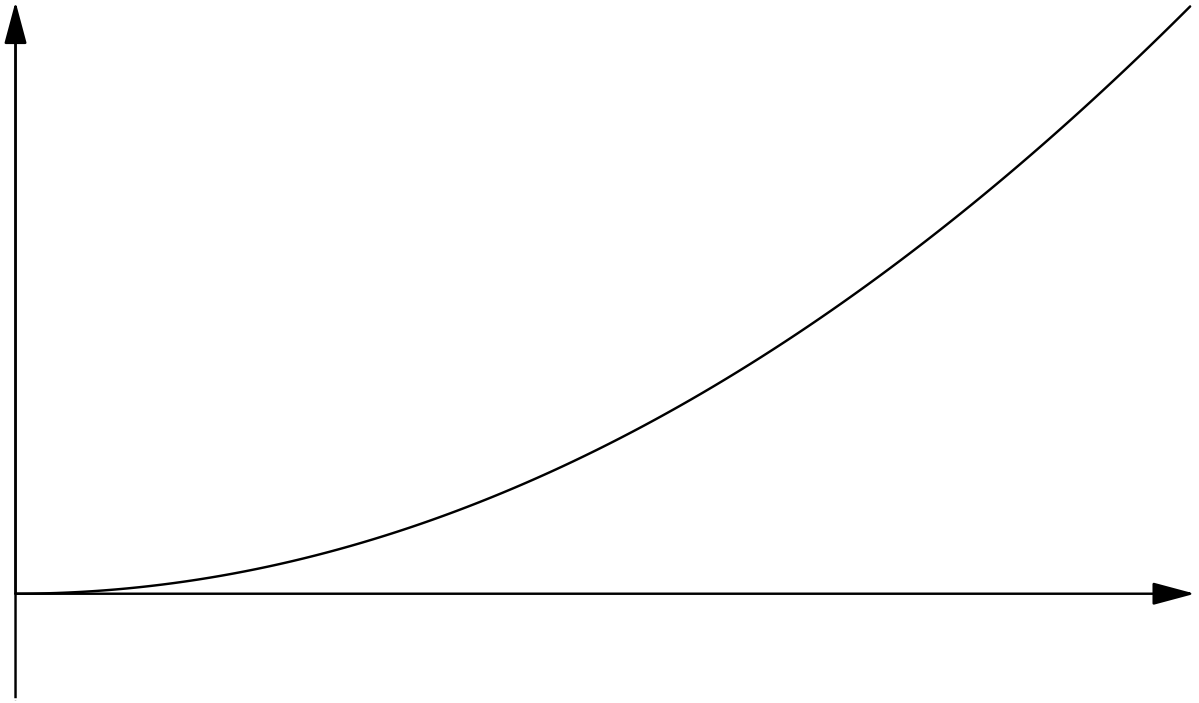
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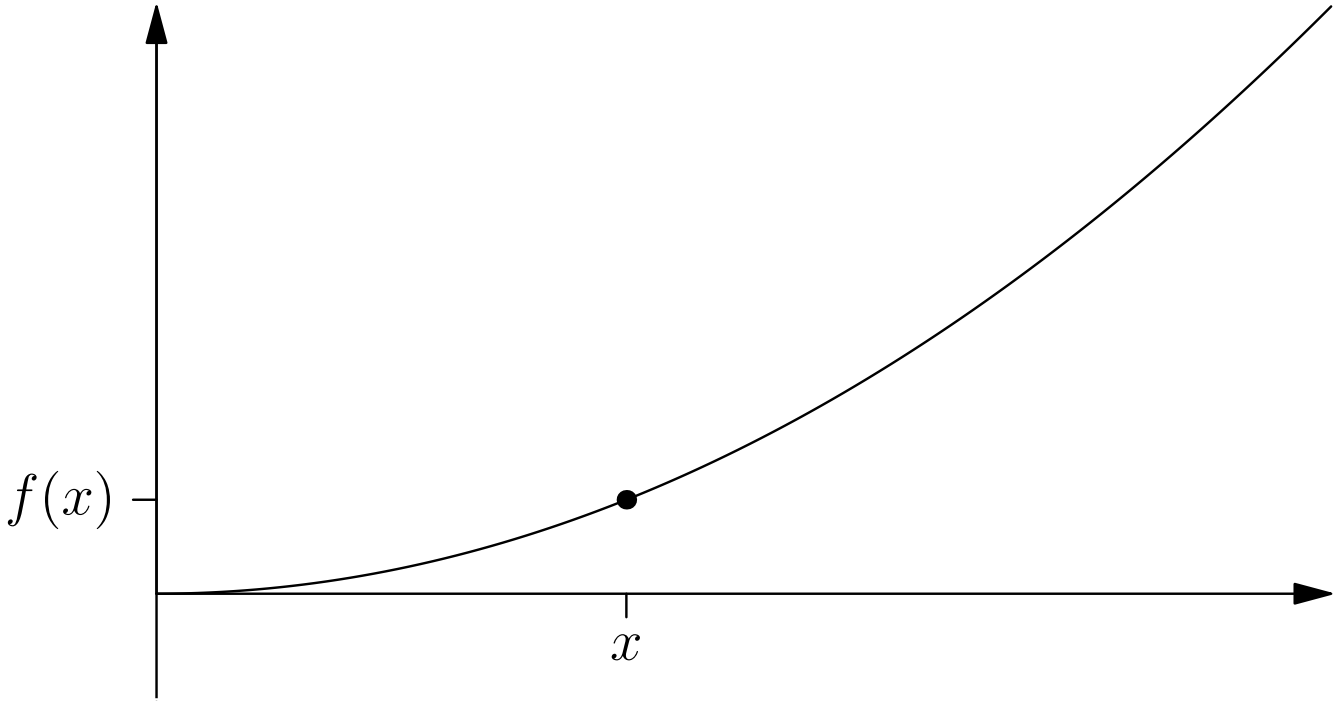
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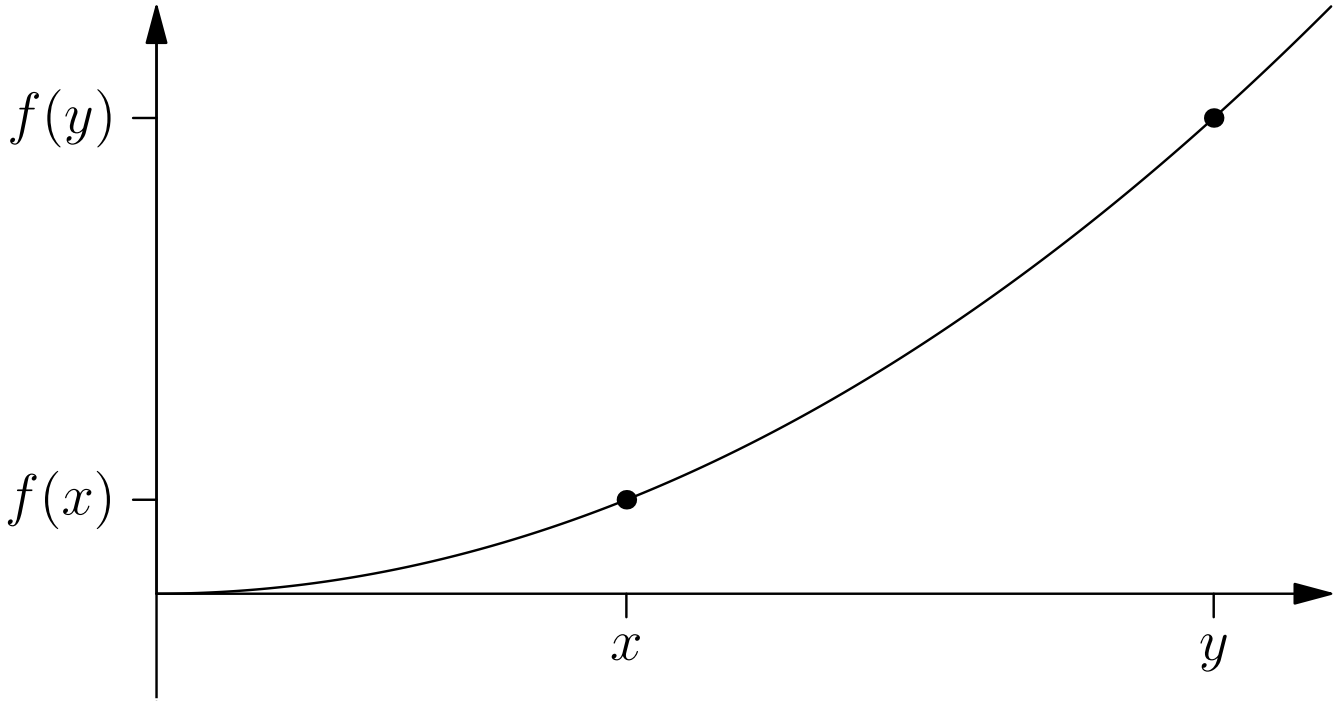
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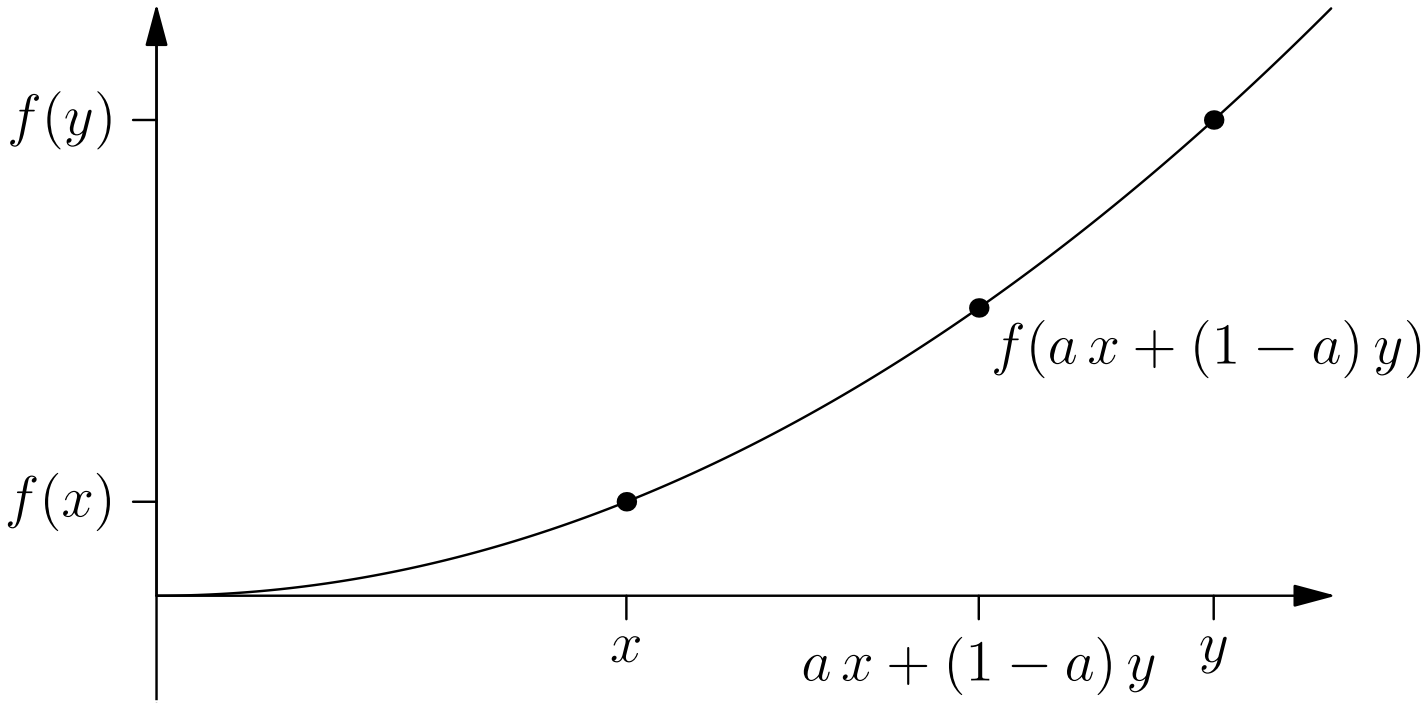
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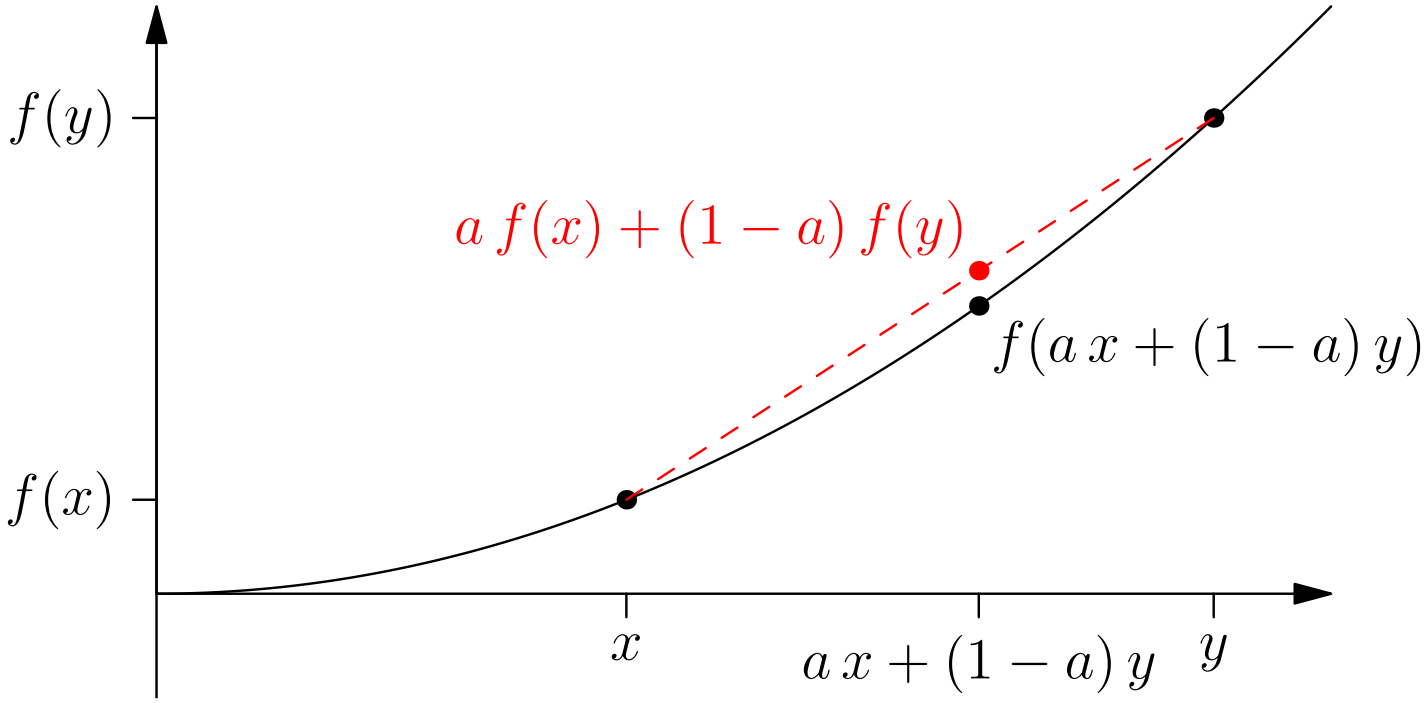
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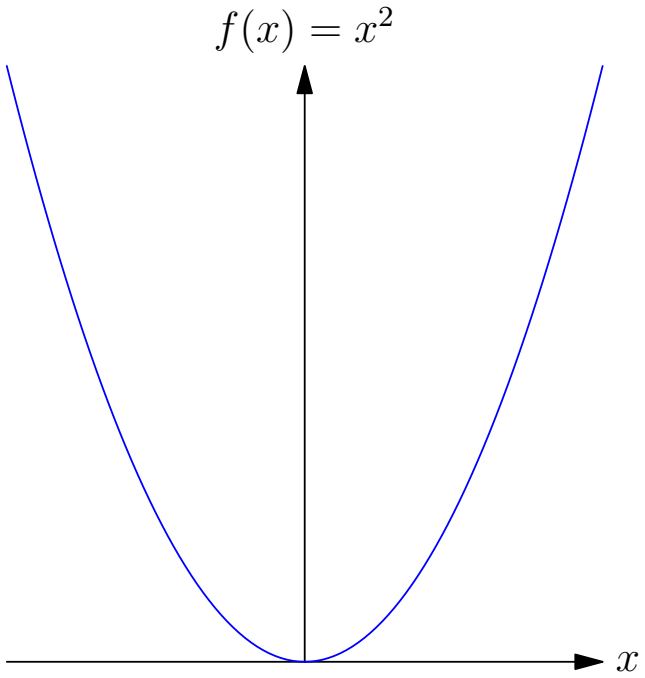
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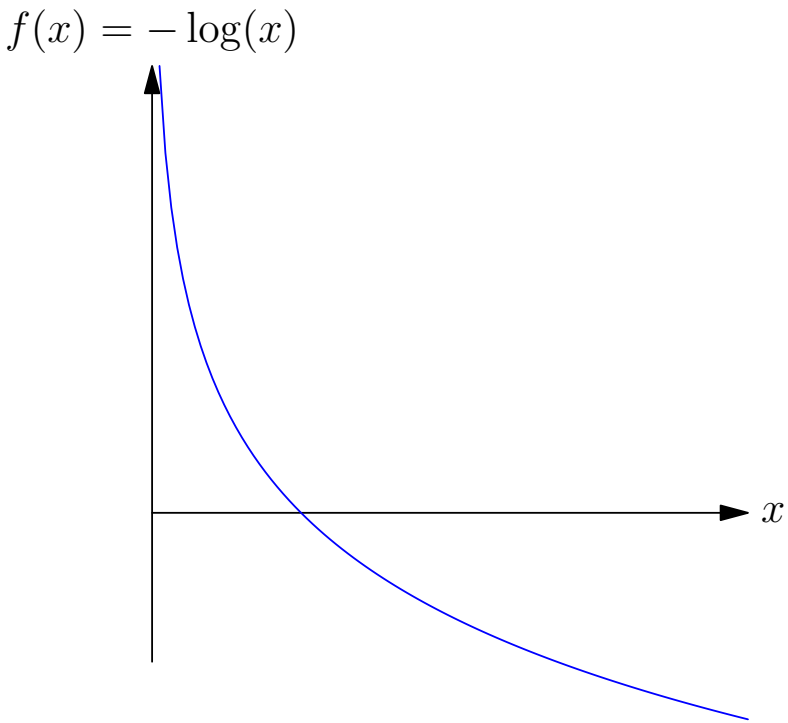


Epigraph

- The **epigraph** of a function is the area that lies above the function

- The epigraph of a convex function is a convex region

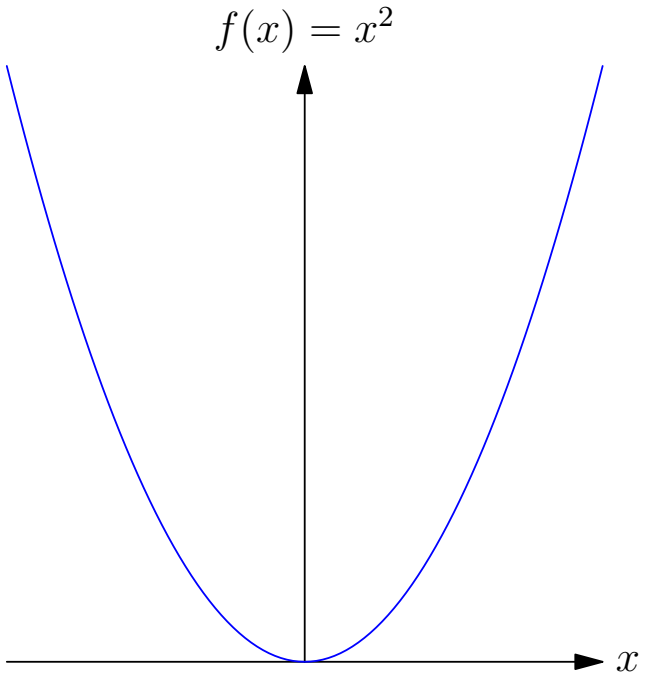


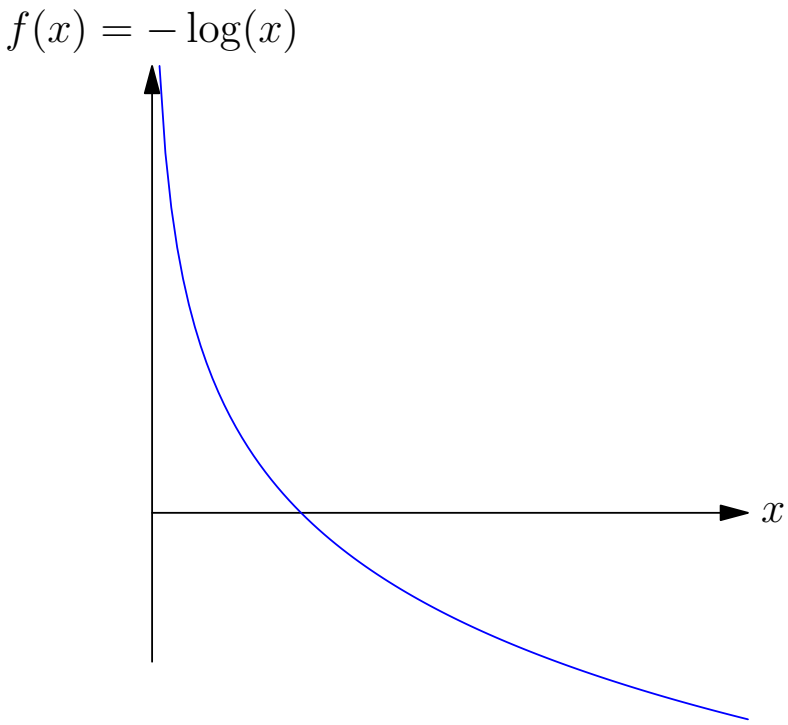


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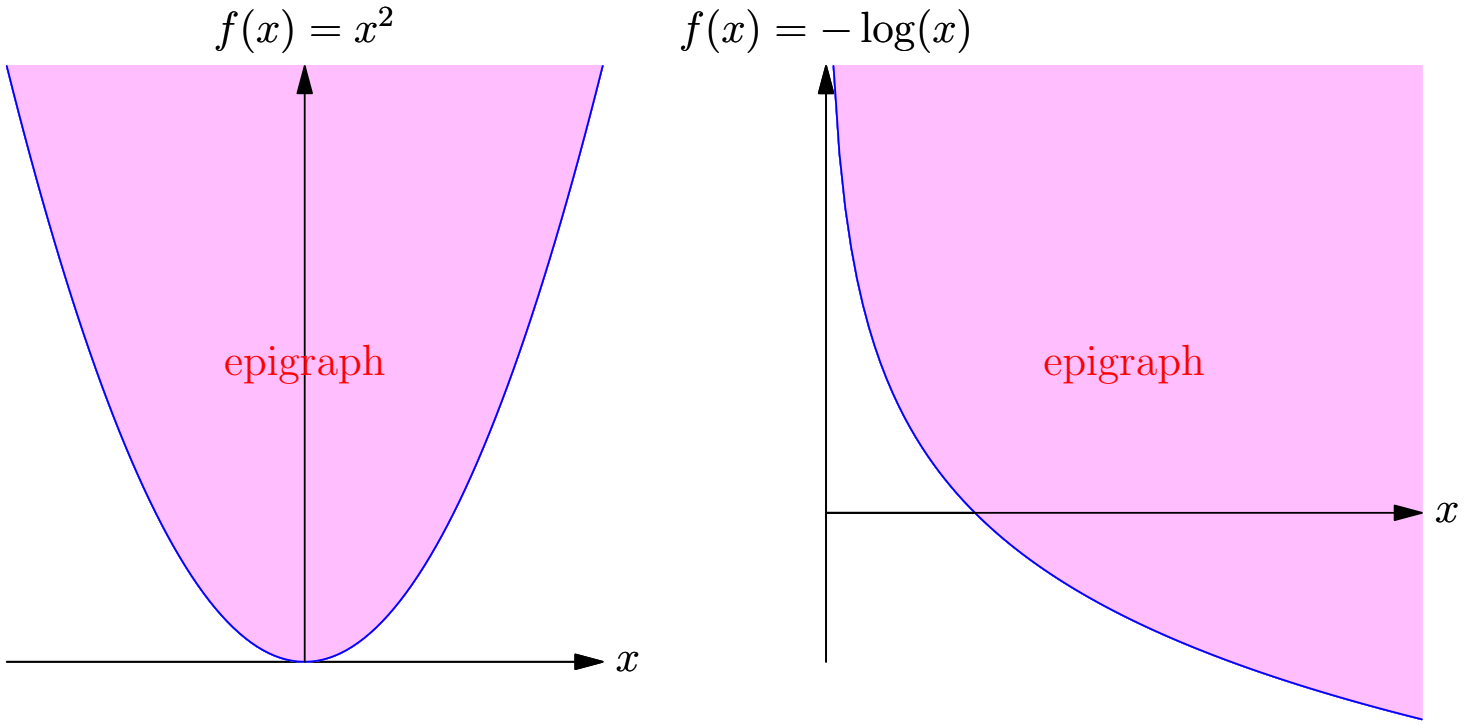




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Convex-Down or Concave Functions

- Any function, $f(x)$, that satisfies the reverse inequality

$$f(ax + (1-a)y) \geq af(x) + (1-a)f(y)$$

  for any points x and y and any $a \in [0,1]$ is said to be a **convex-down** or **concave** function

- Everything true for a convex(-up) function carries over to a convex-down function with a small modification

- If $f(x)$ is a convex-up function then $g(x) = -f(x)$ is a convex-down function

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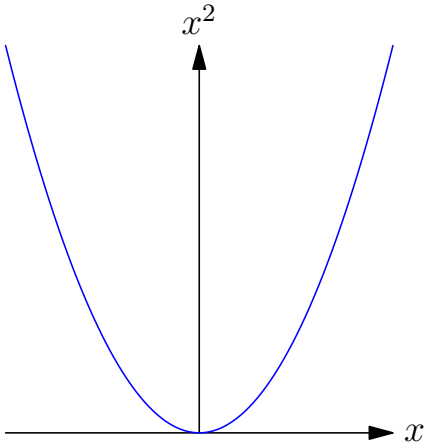
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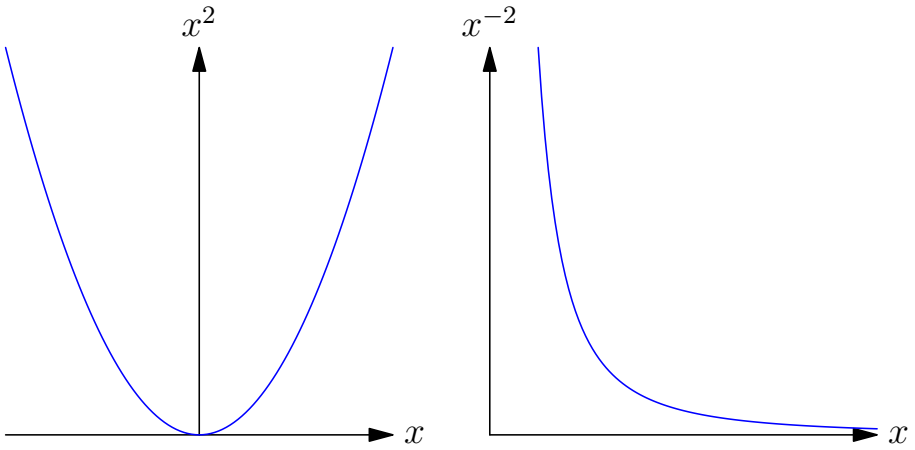
Examples

Convex-Up Functions



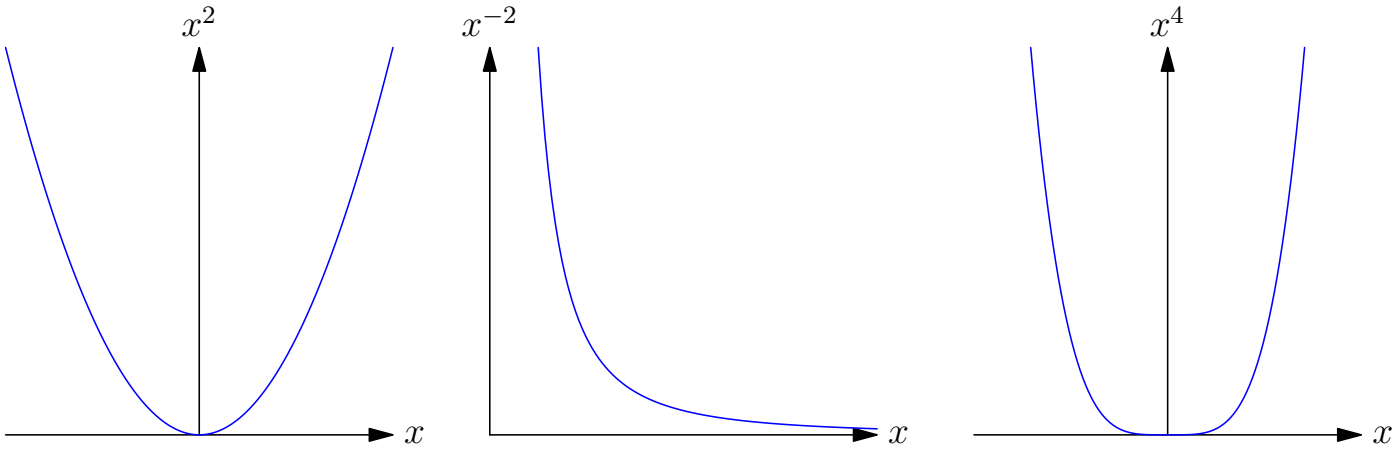
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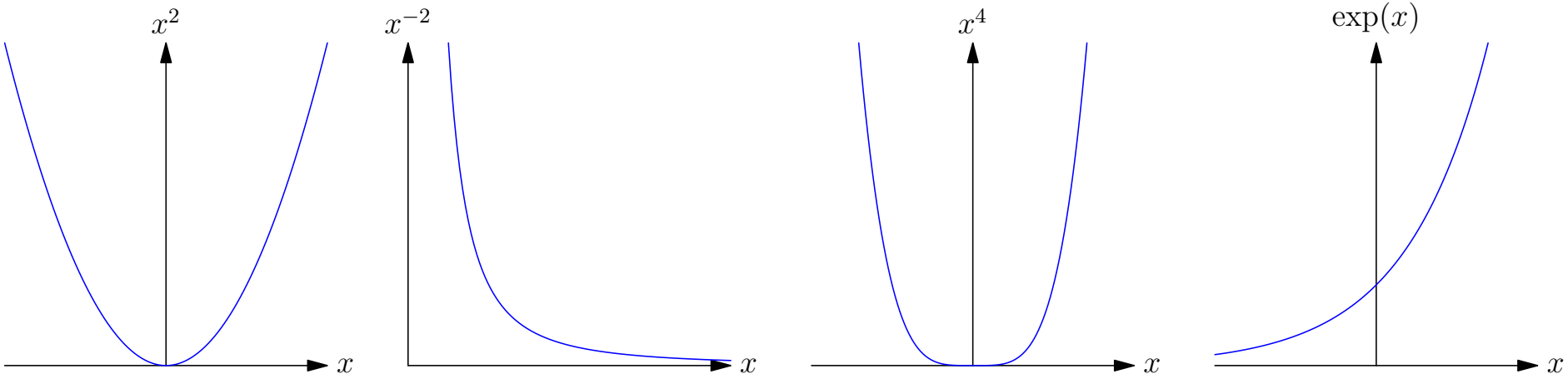
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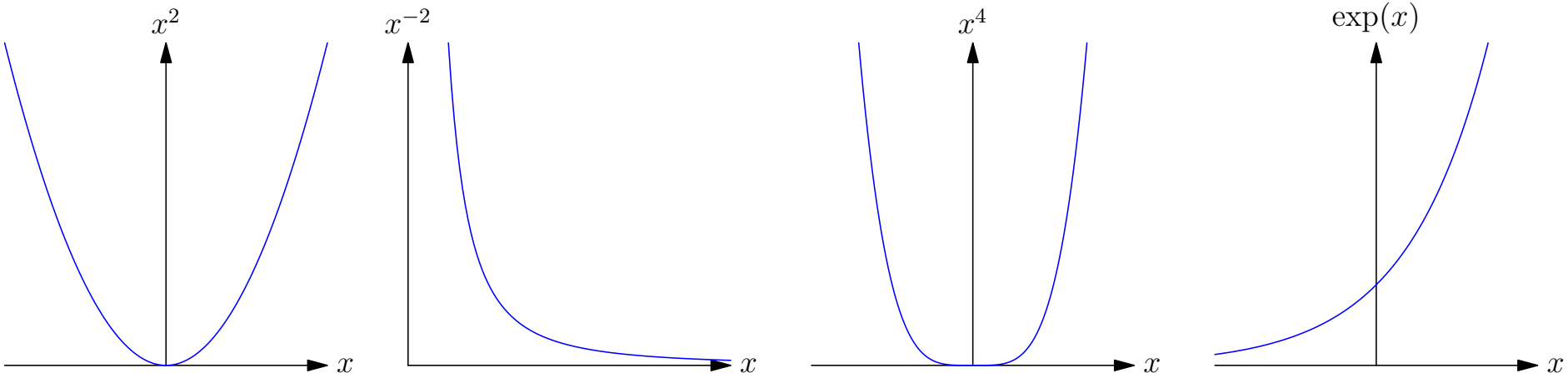
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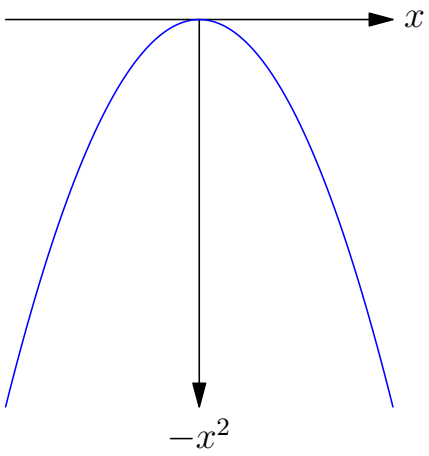


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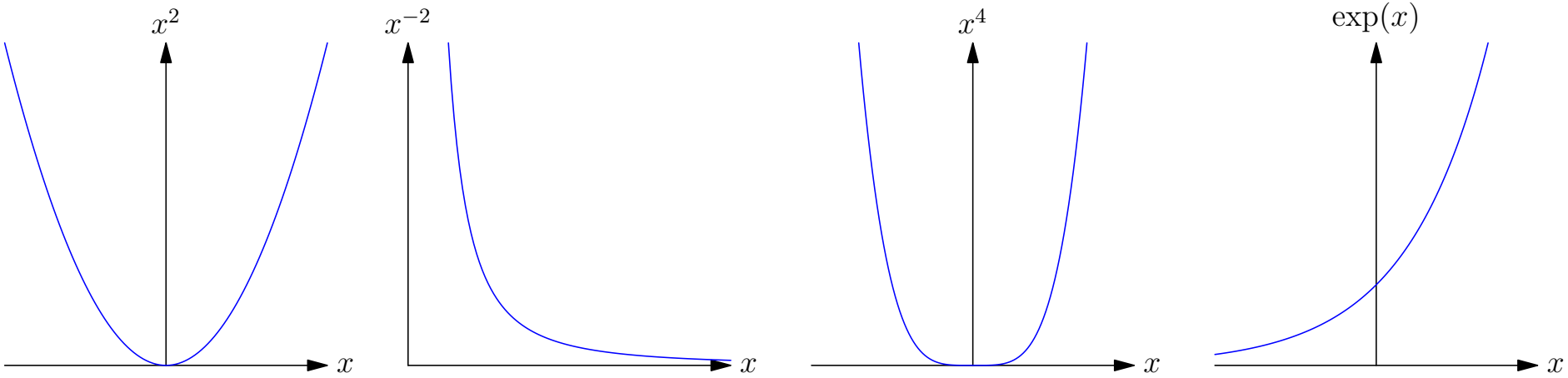


Convex-Down Functions

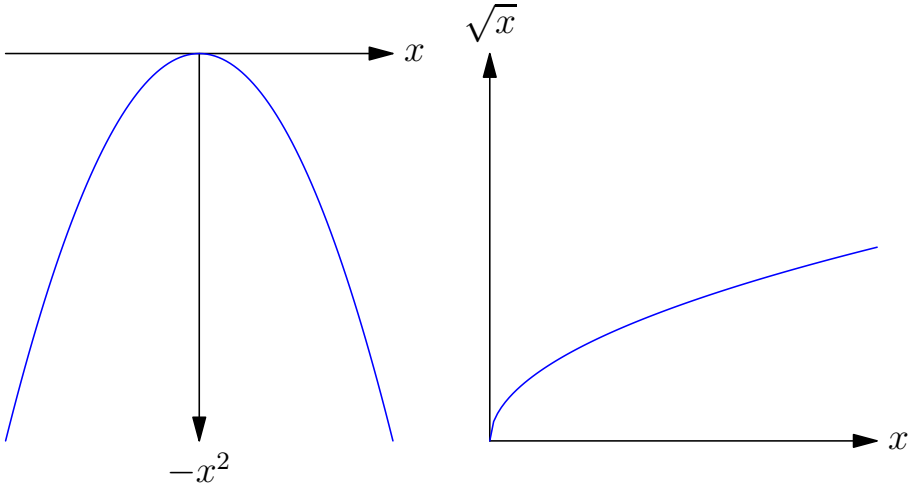


Examples

Convex-Up Functions

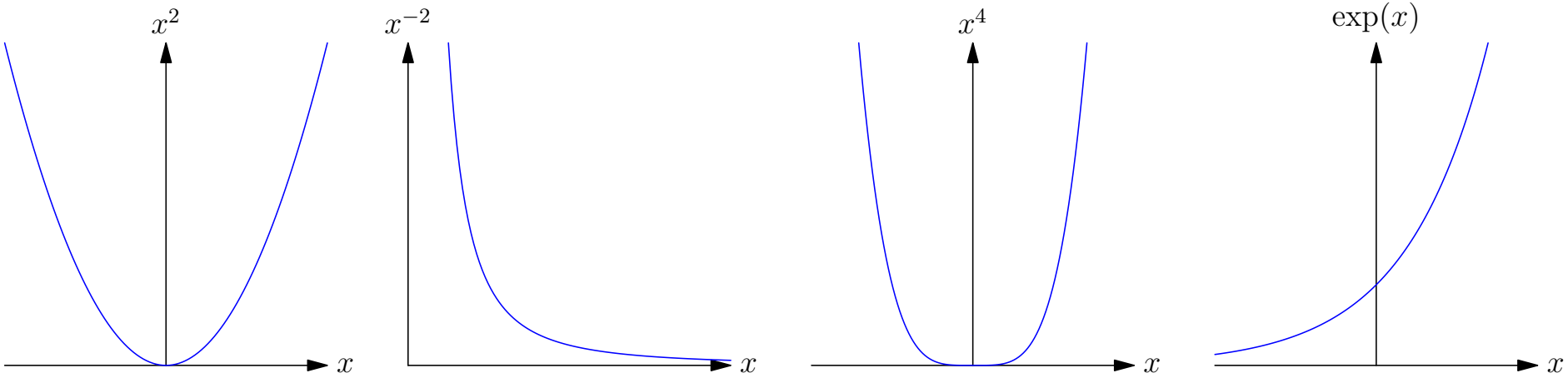


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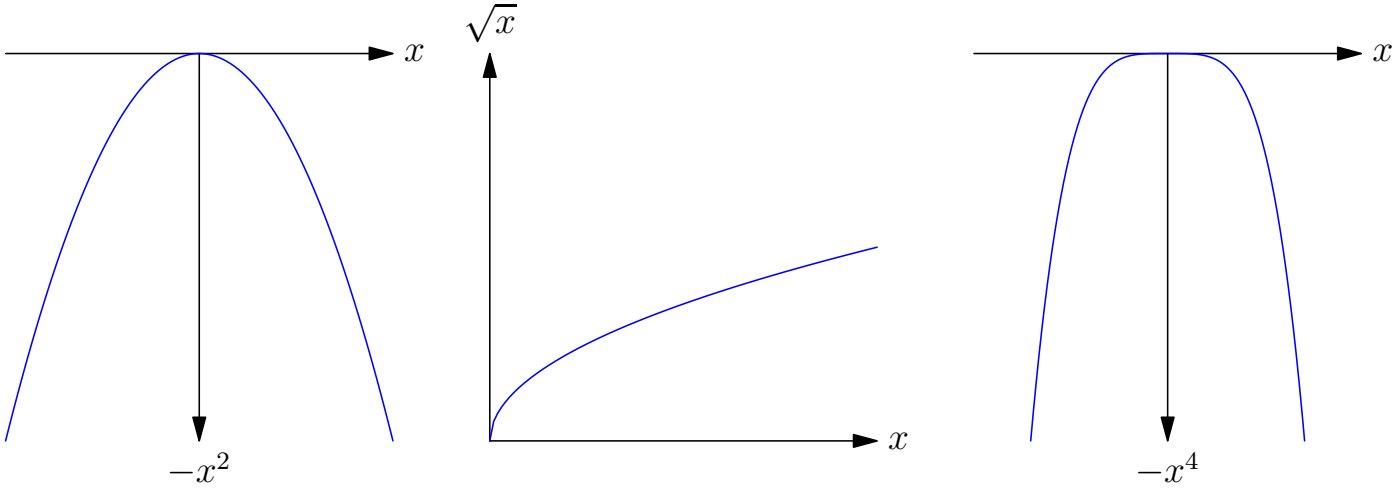


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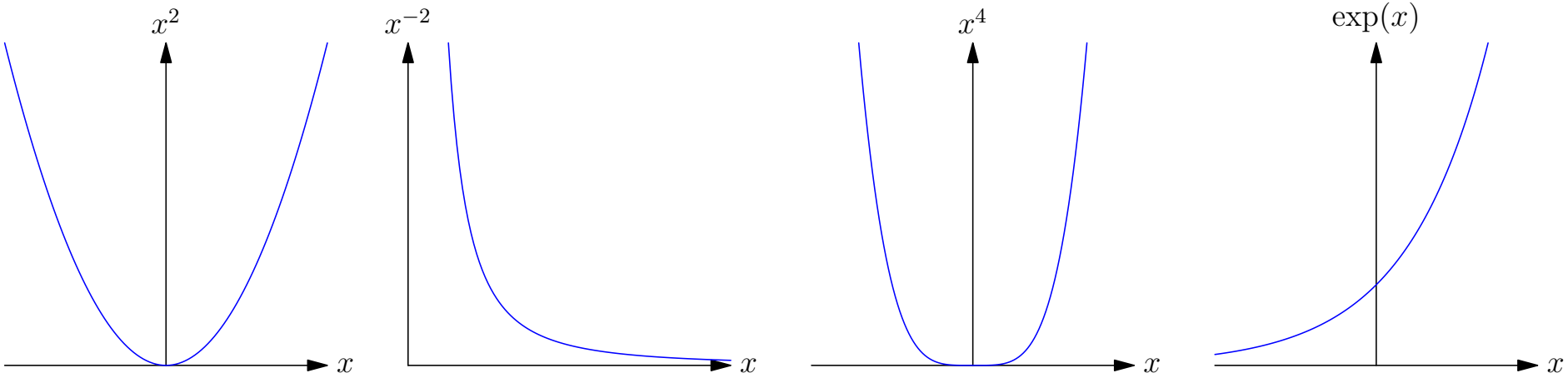


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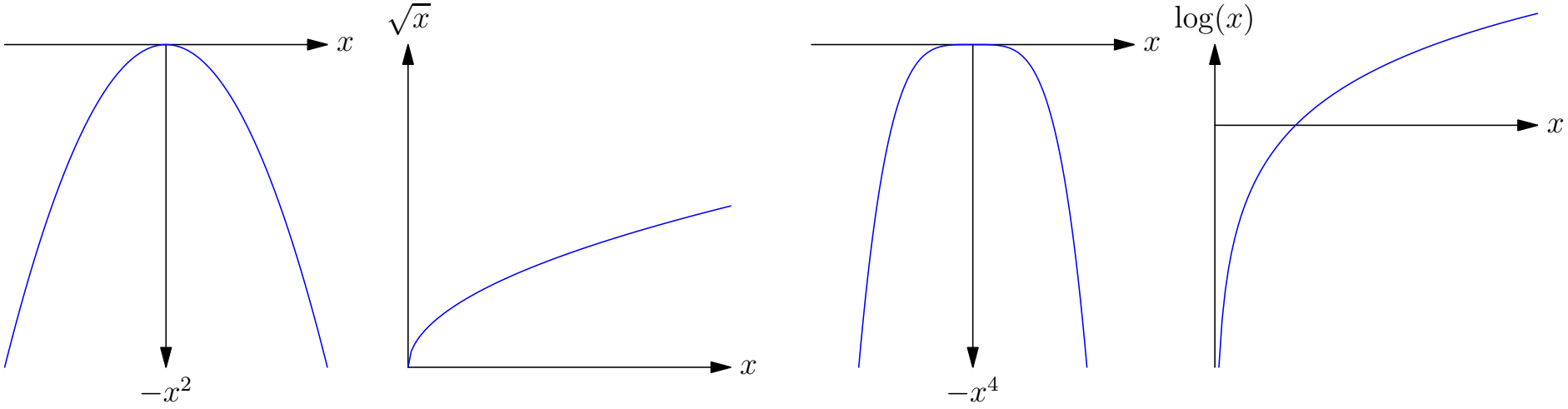


Examples

Convex-Up Functions



Convex-Down Functions



Linear Functions

- Linear functions are given by

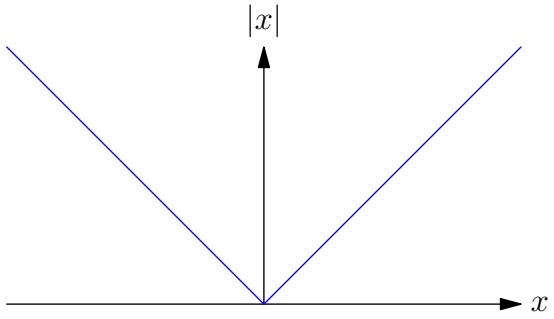
$$f(x) = mx + c$$

- They satisfy the **equality**

$$f(ax + (1-a)y) = af(x) + (1-a)f(y)$$

- As such they are both convex(-up) and convex-down functions

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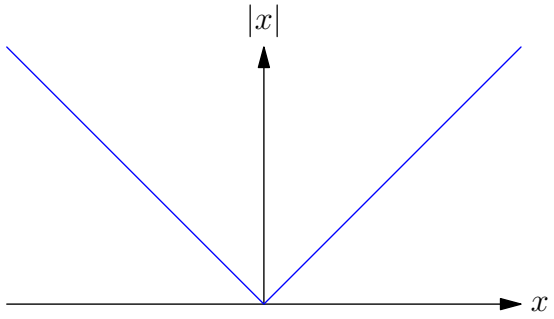
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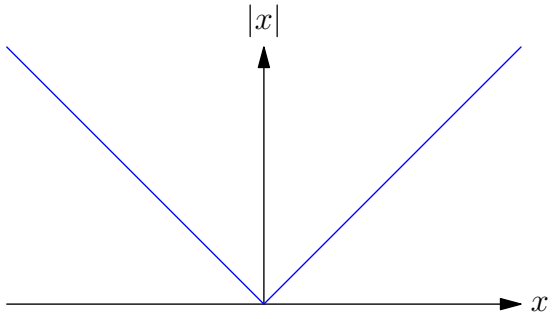
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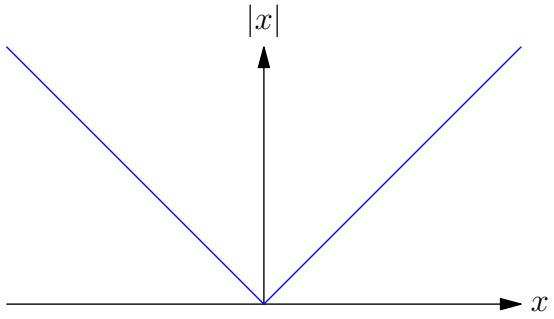
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Strictly Convex Function

- Functions that satisfy the strict inequality (for $0 < a < 1$ and $x \neq y$)

$$f(ax + (1-a)y) < af(x) + (1-a)f(y)$$

  are said to be **strictly convex functions**

- A strictly convex-down function satisfies the reverse strict inequality

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Convexity in High Dimensions

- If $f : \mathbb{R}^n \to \mathbb{R}$ (i.e. $f(\boldsymbol{x})$ maps high dimensional point $\boldsymbol{x} \in \mathbb{R}^n$ to a real value) satisfies

$$f(a\boldsymbol{x} + (1-a)\boldsymbol{y}) \leq a f(\boldsymbol{x}) + (1-a) f(\boldsymbol{y})$$

  for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and any $a \in [0,1]$ then $f(\boldsymbol{x})$ is a convex function

- $\|\boldsymbol{x}\|_2^2 = \sum_i x_i^2$ is a (strictly) convex function

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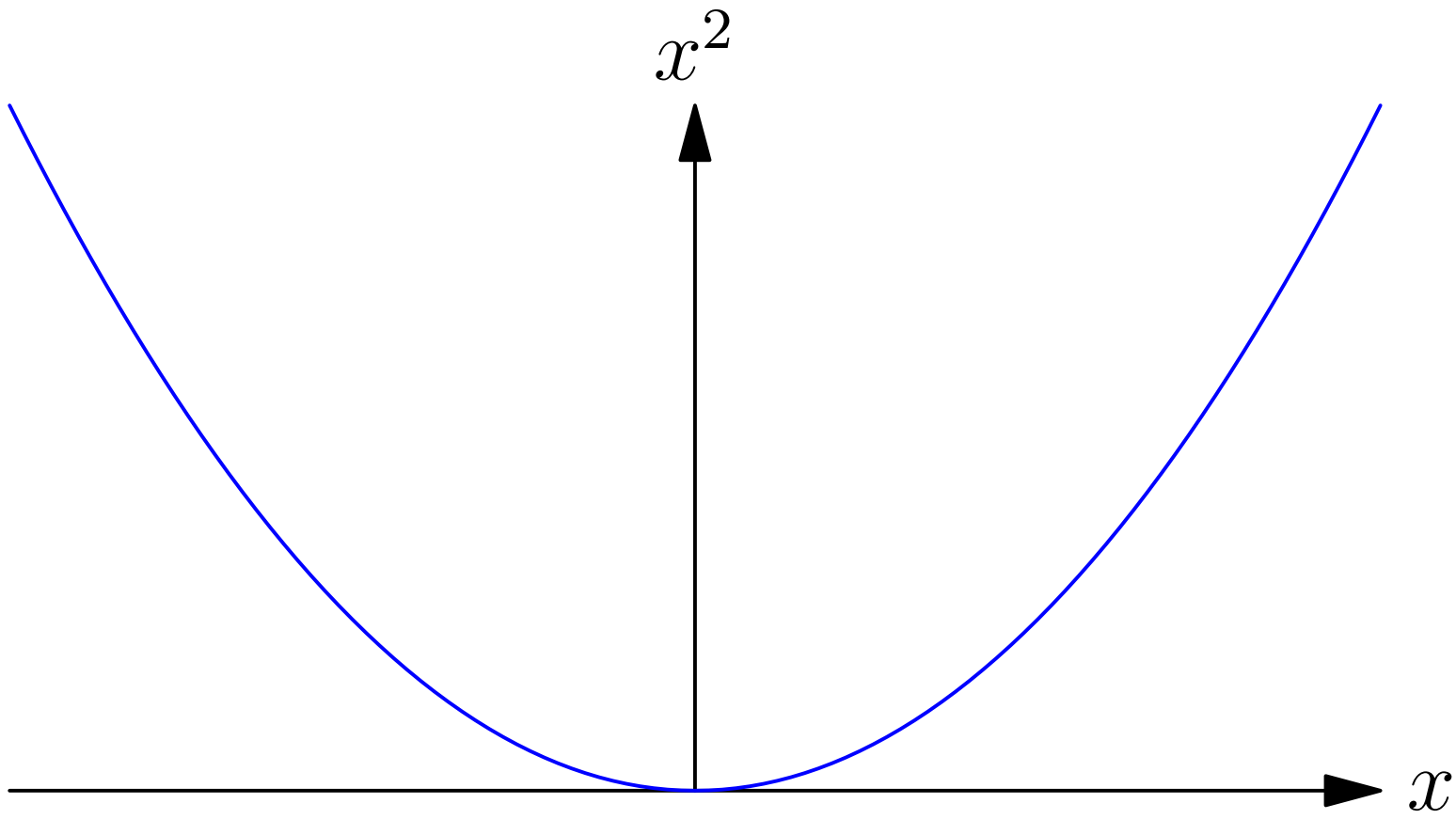
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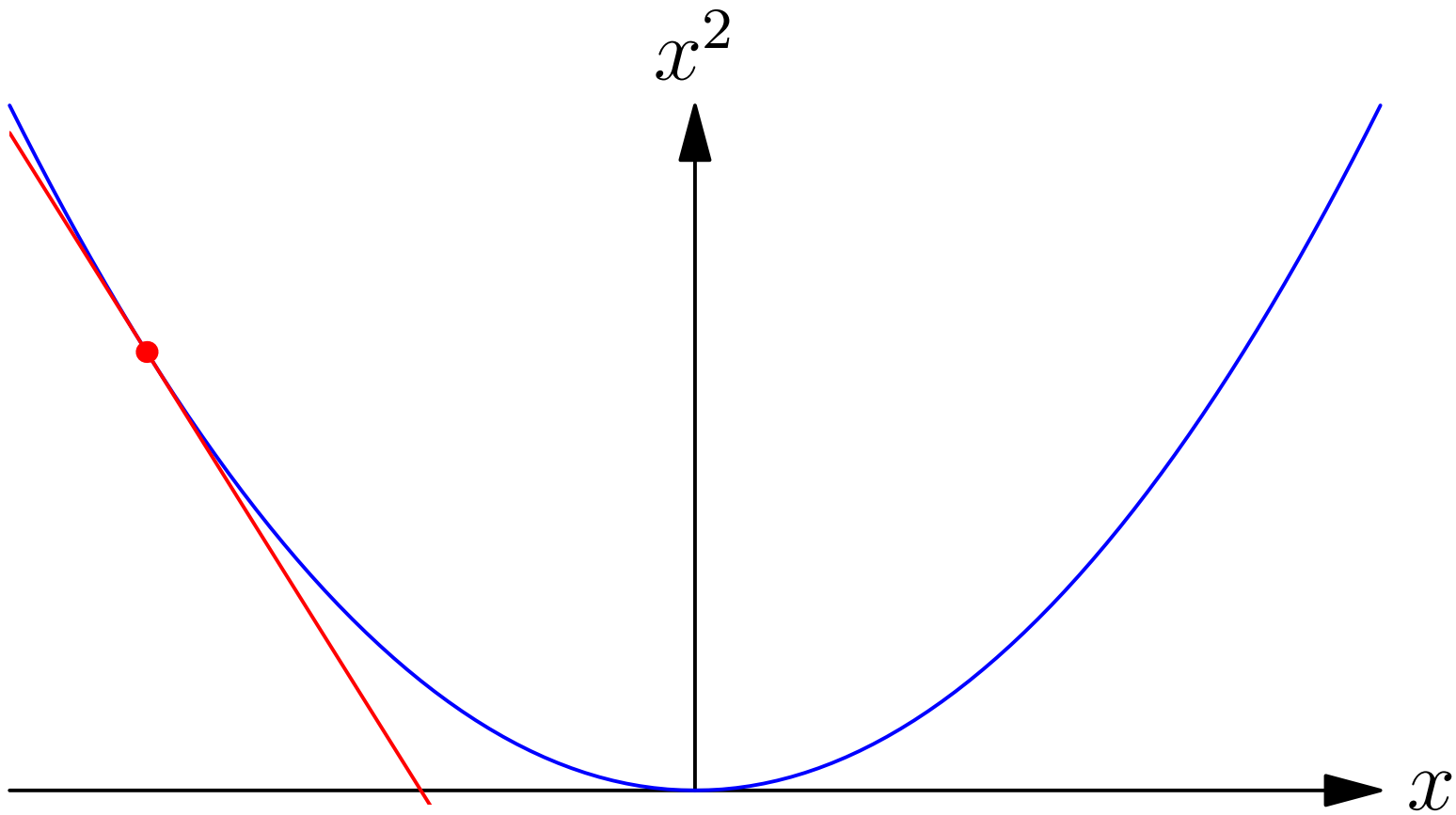
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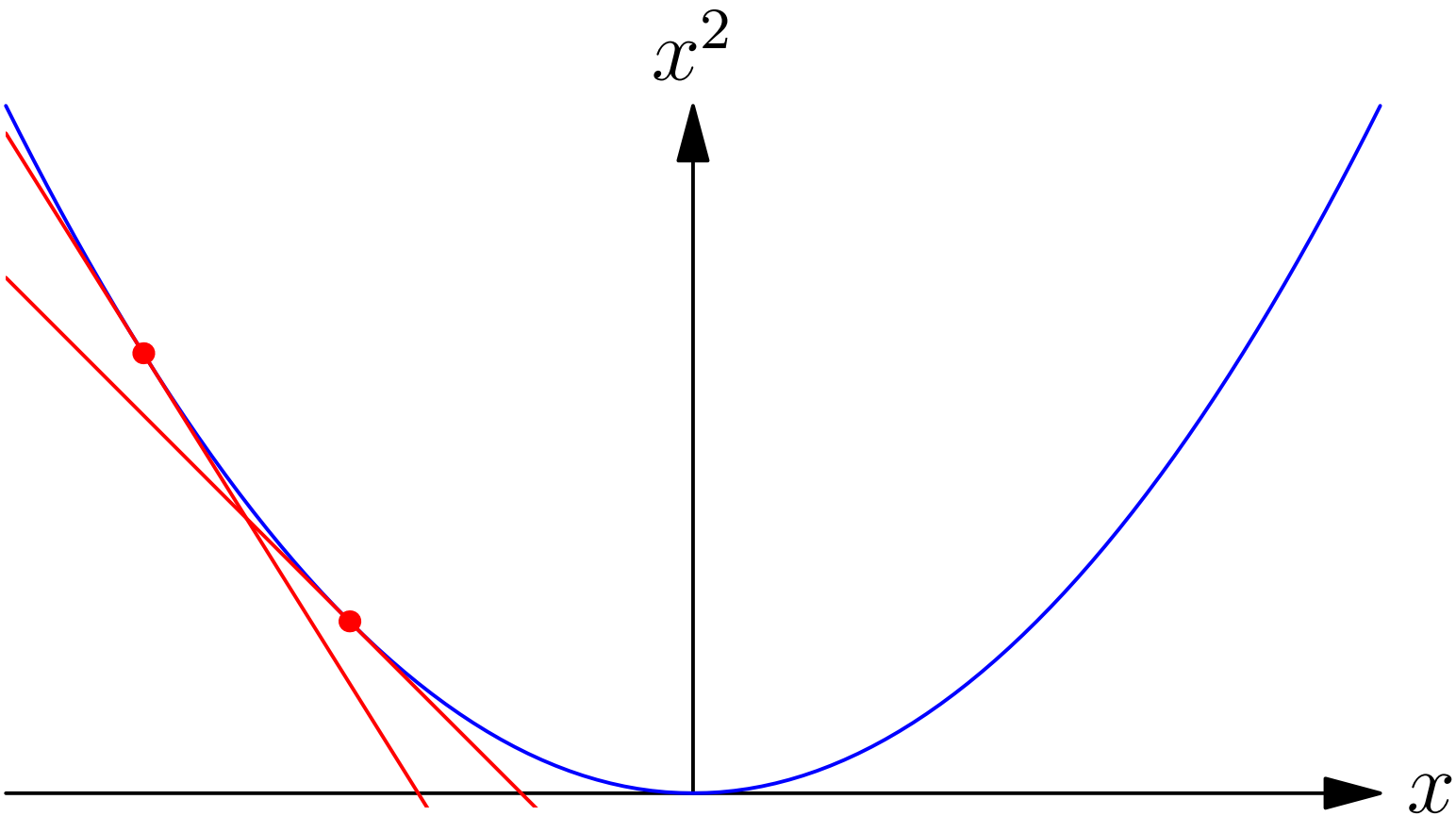
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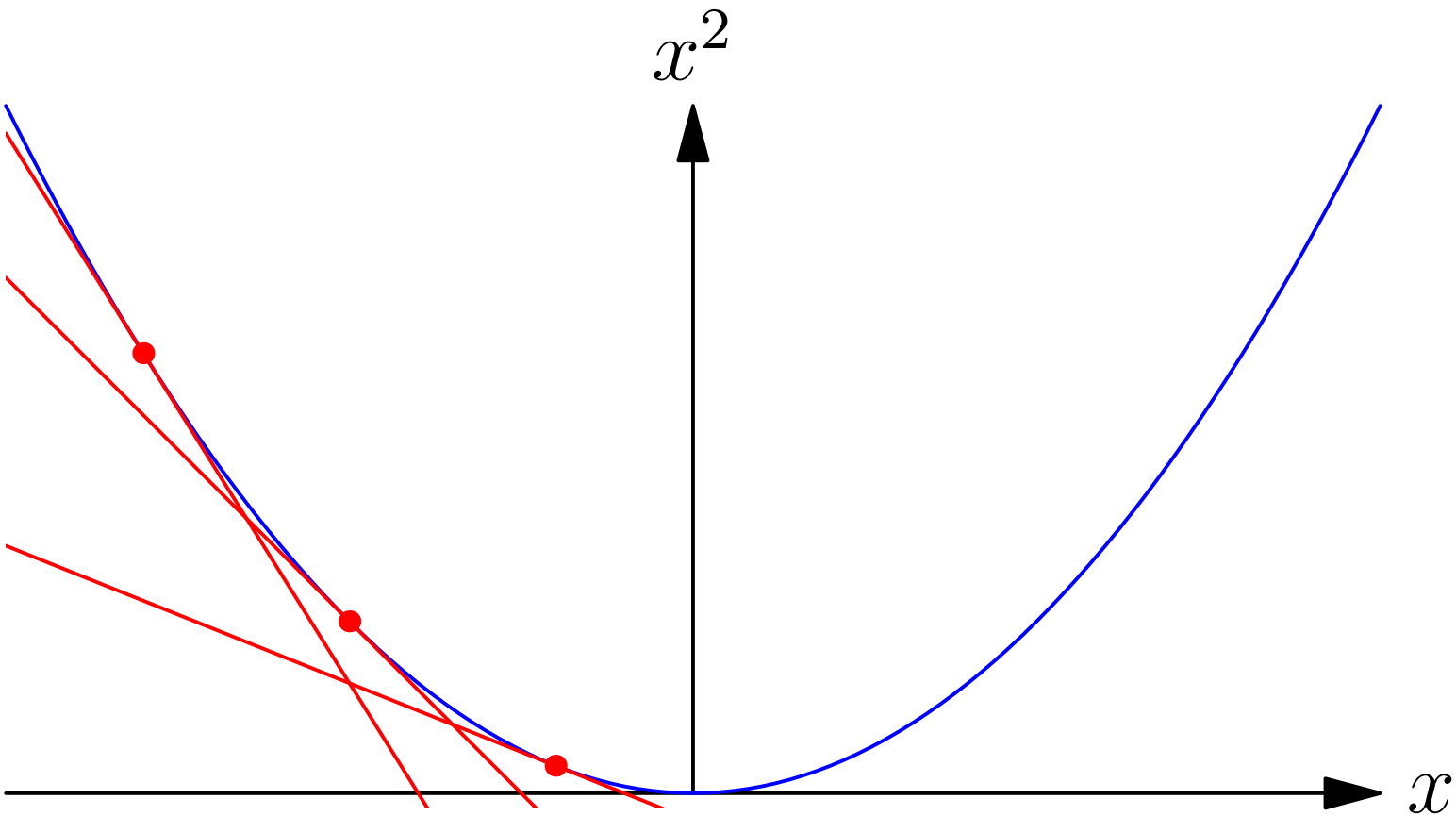
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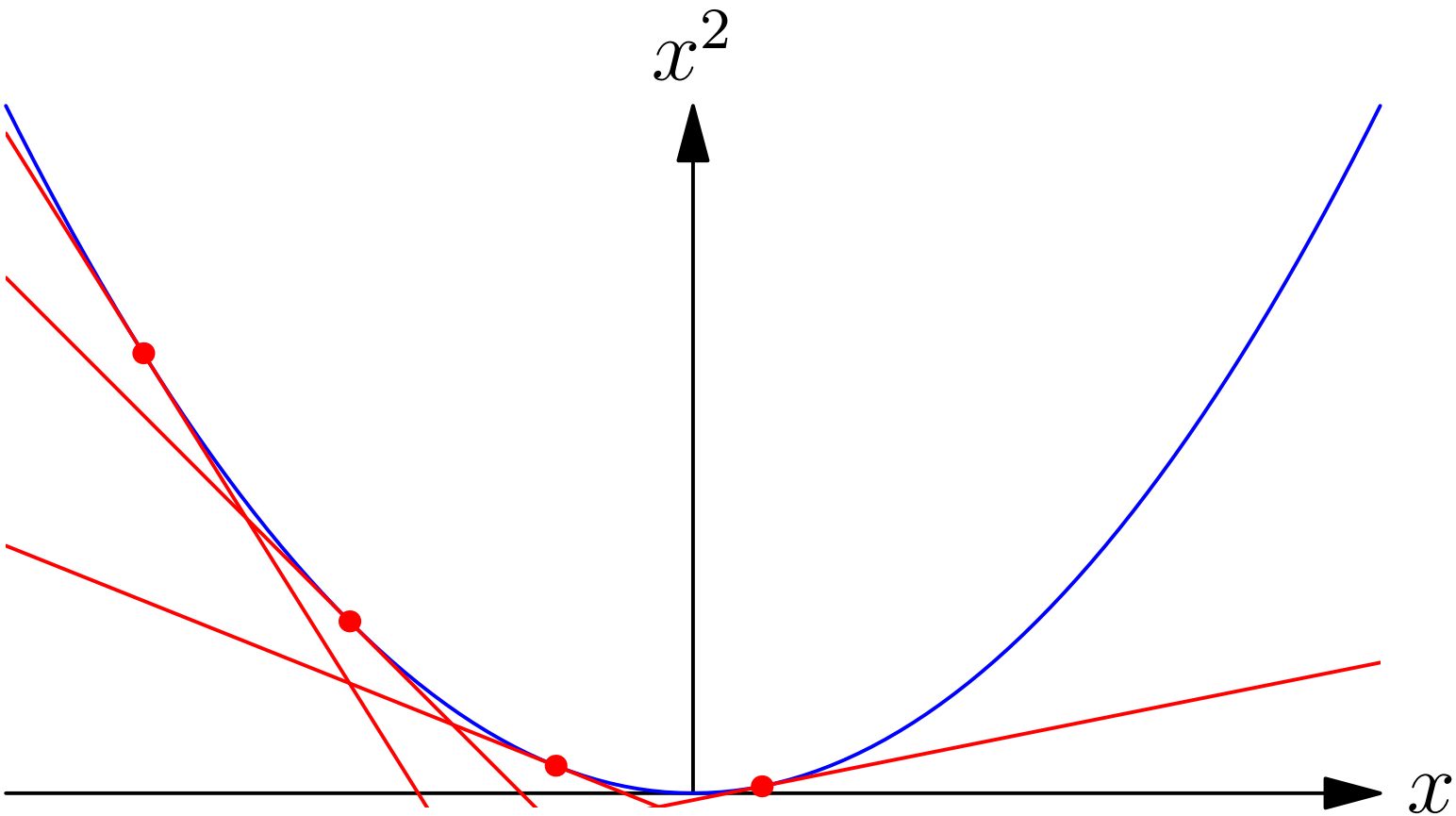
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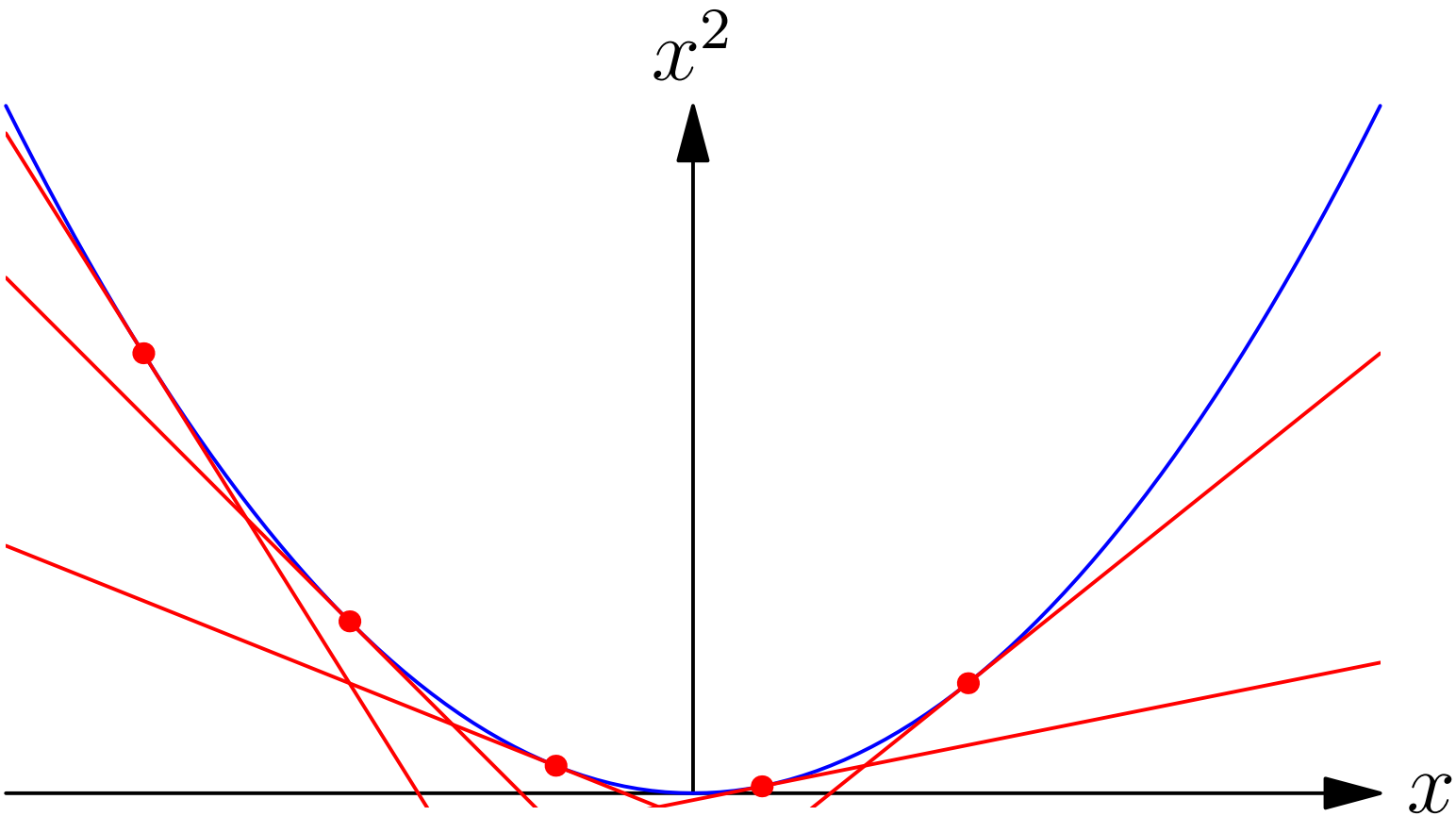
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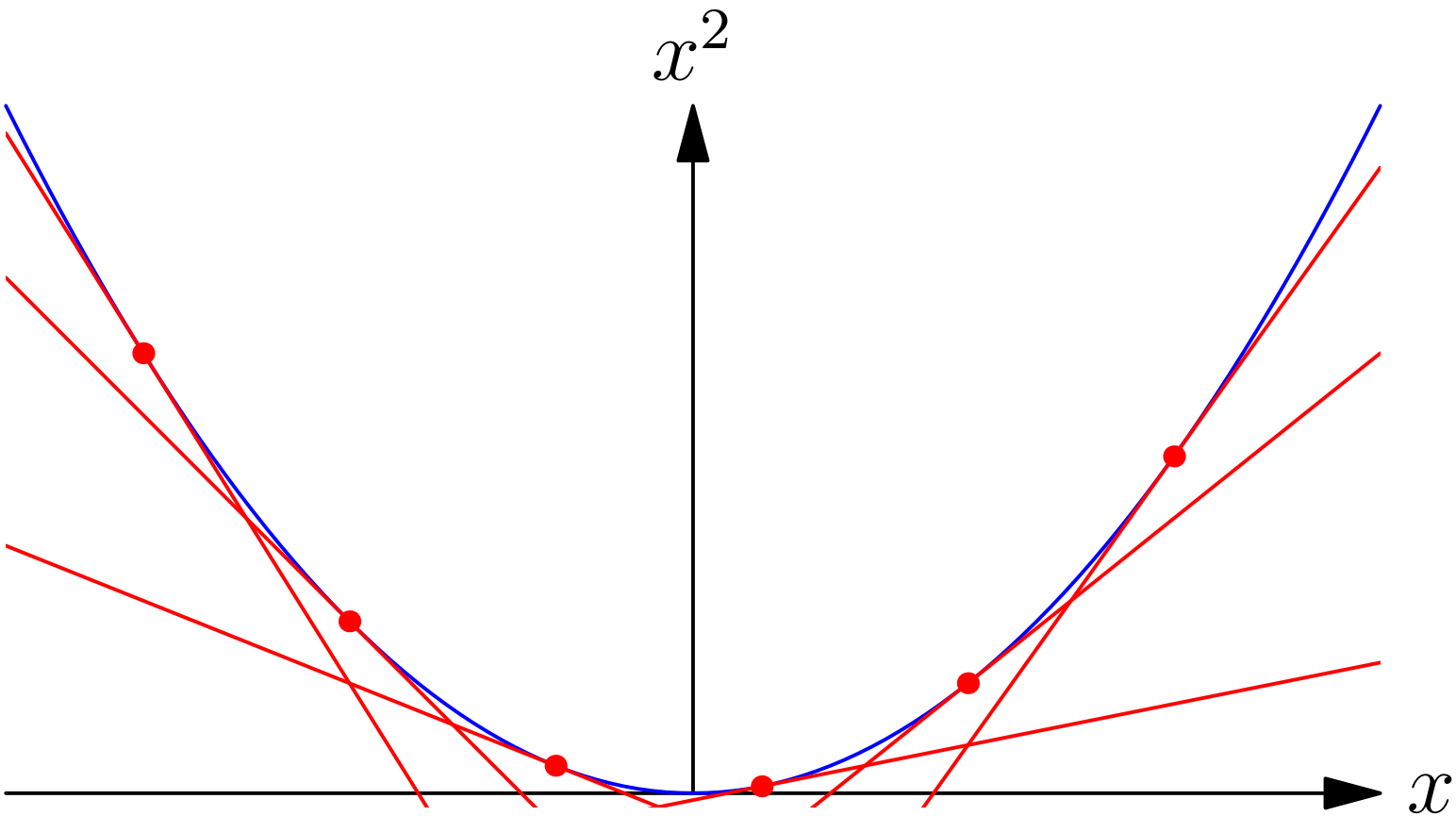
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Second Derivatives

- Adding the tangent-line inequalities at x and $x+\epsilon$, for any $\epsilon > 0$

$$f'(x+\epsilon) \geq f'(x)$$

  therefore $f''(x) = \lim_{\epsilon \to 0}(f'(x+\epsilon) - f'(x))/\epsilon \geq 0$ at all points x

- In high dimensions a convex function lies above its tangent plane

$$f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*) + (\boldsymbol{x} - \boldsymbol{x}^*)^\mathsf{T} \boldsymbol{\nabla} f(\boldsymbol{x}^*)$$

- The matrix of second derivatives (the Hessian) must be positive semi-definite at all points $\boldsymbol{x}$

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Sums of Convex Functions

- For any set of convex functions $f_1(x)$, $f_2(x)$, ... and any set of non-negative scalars a_1, a_2, ... then

$$g(x) = \sum_i a_i f_i(x)$$

  is convex

- Proof

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Convex Functions Defined on Convex Sets

- All the properties we have discussed hold for functions defined on a convex set
- $\sin(x)$ is neither convex-up nor convex-down on the whole real line
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- For a convex function defined on a non-convex set, $\mathcal{S}$, there exist points $\boldsymbol{x},\boldsymbol{y} \in \mathcal{S}$ such that for some $a \in [0,1]$ the point $\boldsymbol{z} = a\boldsymbol{x} + (1-a)\boldsymbol{y} \notin \mathcal{S}$

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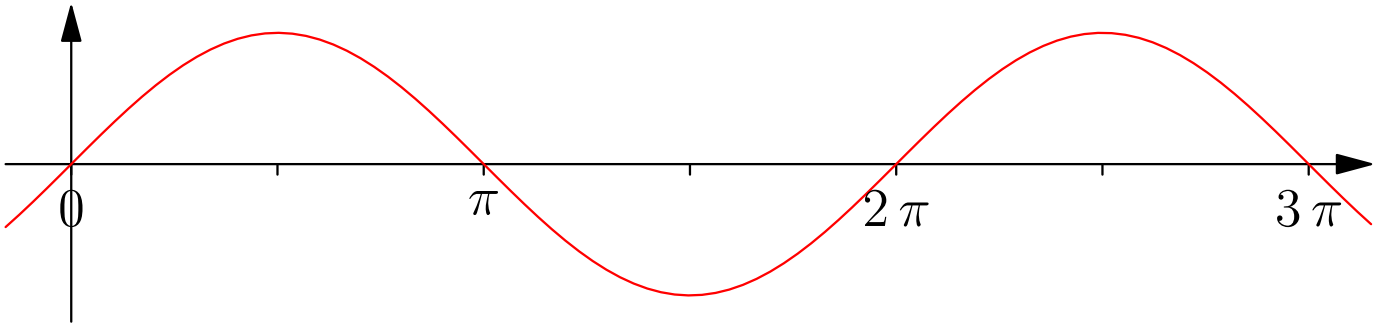
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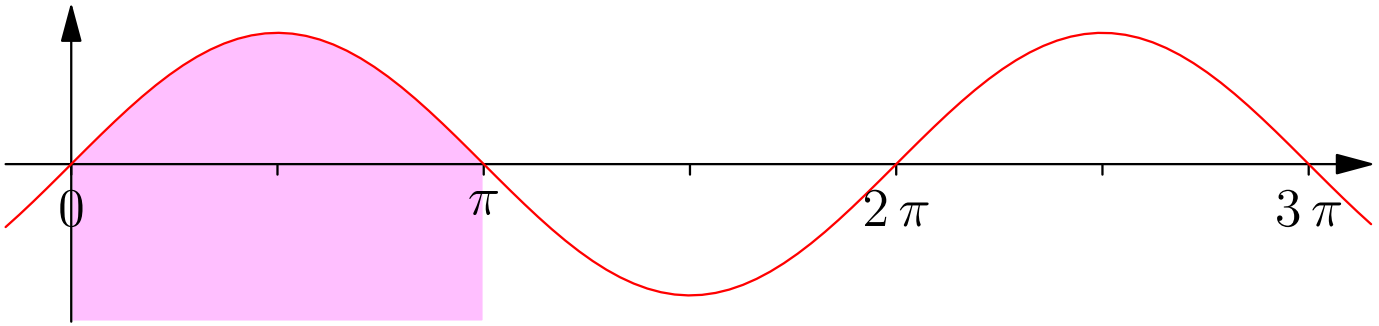
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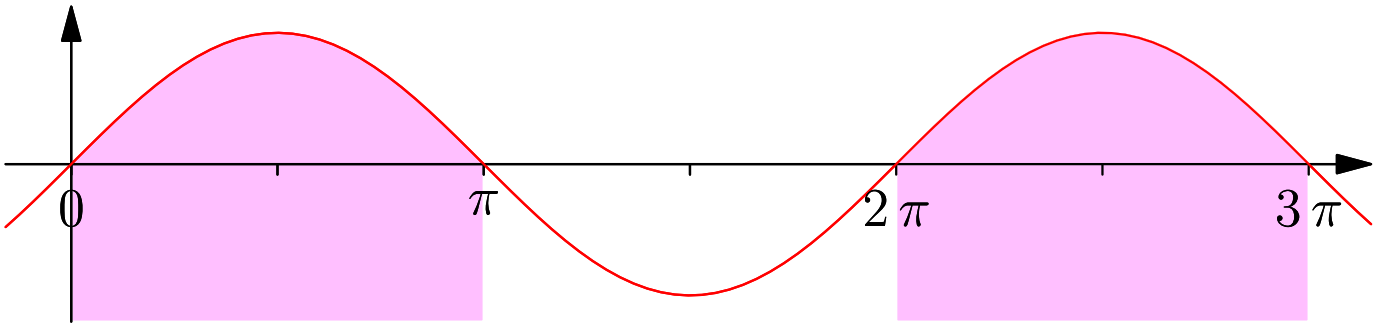
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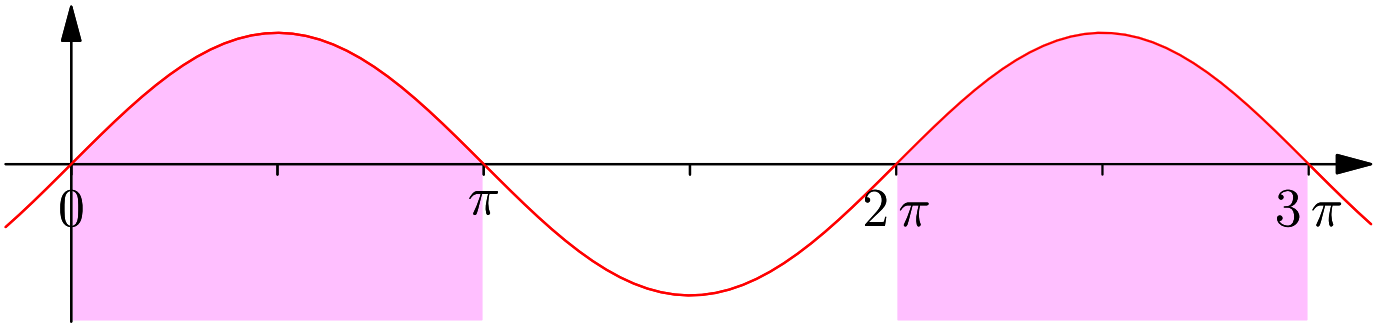
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Constraints

- Often we impose constraints on the set of points, e.g.

$$x_i > 0 \qquad \boldsymbol{a}^\mathsf{T}\boldsymbol{x} = b \qquad \boldsymbol{x}^\mathsf{T}\boldsymbol{M}\boldsymbol{x} \leq 1$$

- Linear constraints (e.g. $x_i > 0$ or $\boldsymbol{a}^\mathsf{T}\boldsymbol{x} = b$ or $\boldsymbol{a}^\mathsf{T}\boldsymbol{x} \leq b$) always define a convex region
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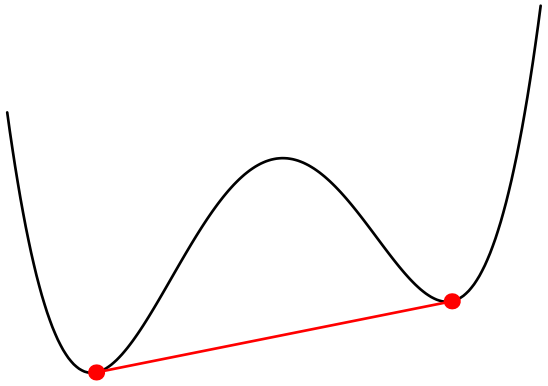
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Unique Minimum

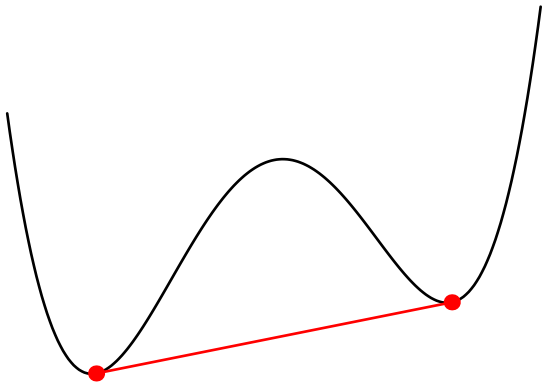
- A strictly convex function has at most one minimum (e^x has none)
- A local minimum that is not global would break convexity
  - ⋆ On the line from a local minimum to a lower global minimum f lies on or below the chord, which strictly decreases
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- This remains true if we consider convex functions that are constrained to live in a convex set

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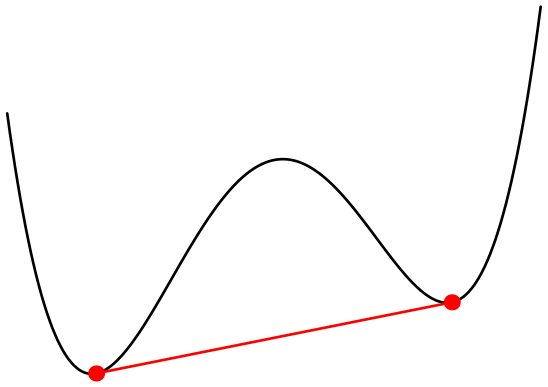
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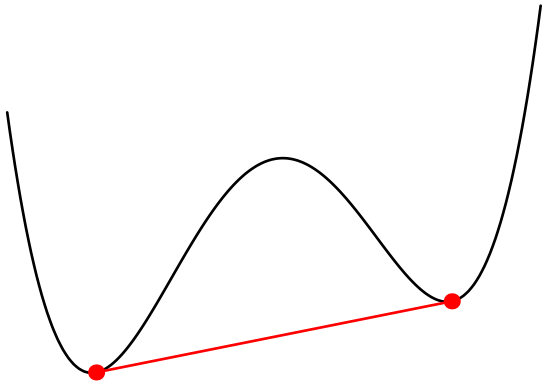
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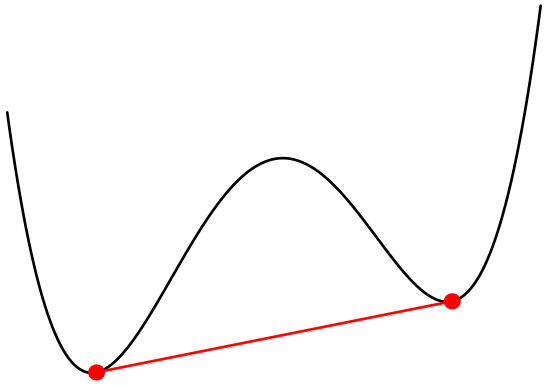
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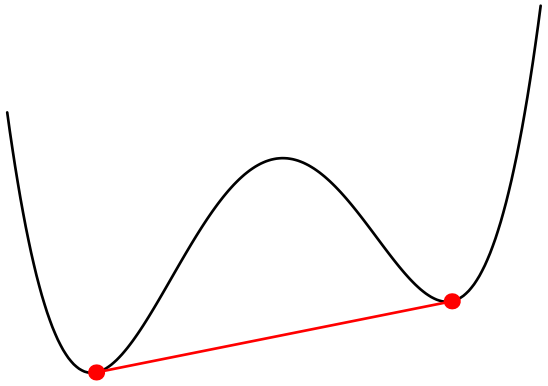
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Convex Set of Minima

- If $f(\boldsymbol{x})$ is **convex** but not **strictly convex** then there might exist a convex set $\mathcal{M} \subset \mathcal{X}$ of minima such that for all $\boldsymbol{x},\boldsymbol{y} \in \mathcal{M}$ and any $\boldsymbol{z} \in \mathcal{X}$ we have $f(\boldsymbol{x}) = f(\boldsymbol{y}) \leq f(\boldsymbol{z})$

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Linear Regression

- For linear regression the loss function

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- If $\mathbf{H} \succ 0$ there will be a unique minimum, while if $\mathbf{H}$ has some zero eigenvalues there will be a family of solutions

Regularised Linear Regression

- In ridge regression we minimise a loss

$$L(\boldsymbol{w}) = \|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2 + \eta\|\boldsymbol{w}\|^2 = \boldsymbol{w}^\mathsf{T}\left(\mathbf{X}^\mathsf{T}\mathbf{X} + \eta\mathbf{I}\right)\boldsymbol{w} - 2\boldsymbol{w}^\mathsf{T}\mathbf{X}^\mathsf{T}\boldsymbol{y} + \boldsymbol{y}^\mathsf{T}\boldsymbol{y}$$

- Because $\|\boldsymbol{w}\|^2$ is strictly convex the loss function is strictly convex and so will have a unique solution

- Using an L_1 regulariser (Lasso)

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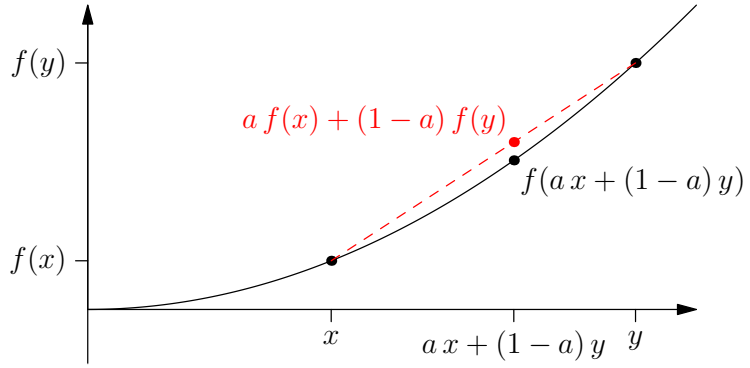
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Outline

1. Convex sets
2. Convex functions
3. **Jensen's inequality**



Jensen's Inequality

- In proving many properties of learning machines inequalities are really useful
- One of the most useful inequalities involves expectations of convex functions, this is known as **Jensen's Inequality**
- If $f(\boldsymbol{x})$ is a convex(-up) function then

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Proof

- We said before that a (differentiable) convex function must lie on or above its tangent plane at any point $\boldsymbol{x}^*$

$$f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*) + (\boldsymbol{x} - \boldsymbol{x}^*)^\mathsf{T} \boldsymbol{\nabla} f(\boldsymbol{x}^*)$$

- Taking $\boldsymbol{x}^* = \mathbb{E}[\boldsymbol{X}]$

$$f(\boldsymbol{X}) \geq f(\mathbb{E}[\boldsymbol{X}]) + (\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}])^\mathsf{T} \boldsymbol{\nabla} f(\mathbb{E}[\boldsymbol{X}])$$

- Taking expectations of both sides

$$\mathbb{E}[f(\boldsymbol{X})] \geq f(\mathbb{E}[\boldsymbol{X}]) + (\mathbb{E}[\boldsymbol{X}] - \mathbb{E}[\boldsymbol{X}])^\mathsf{T} \boldsymbol{\nabla} f(\mathbb{E}[\boldsymbol{X}])$$

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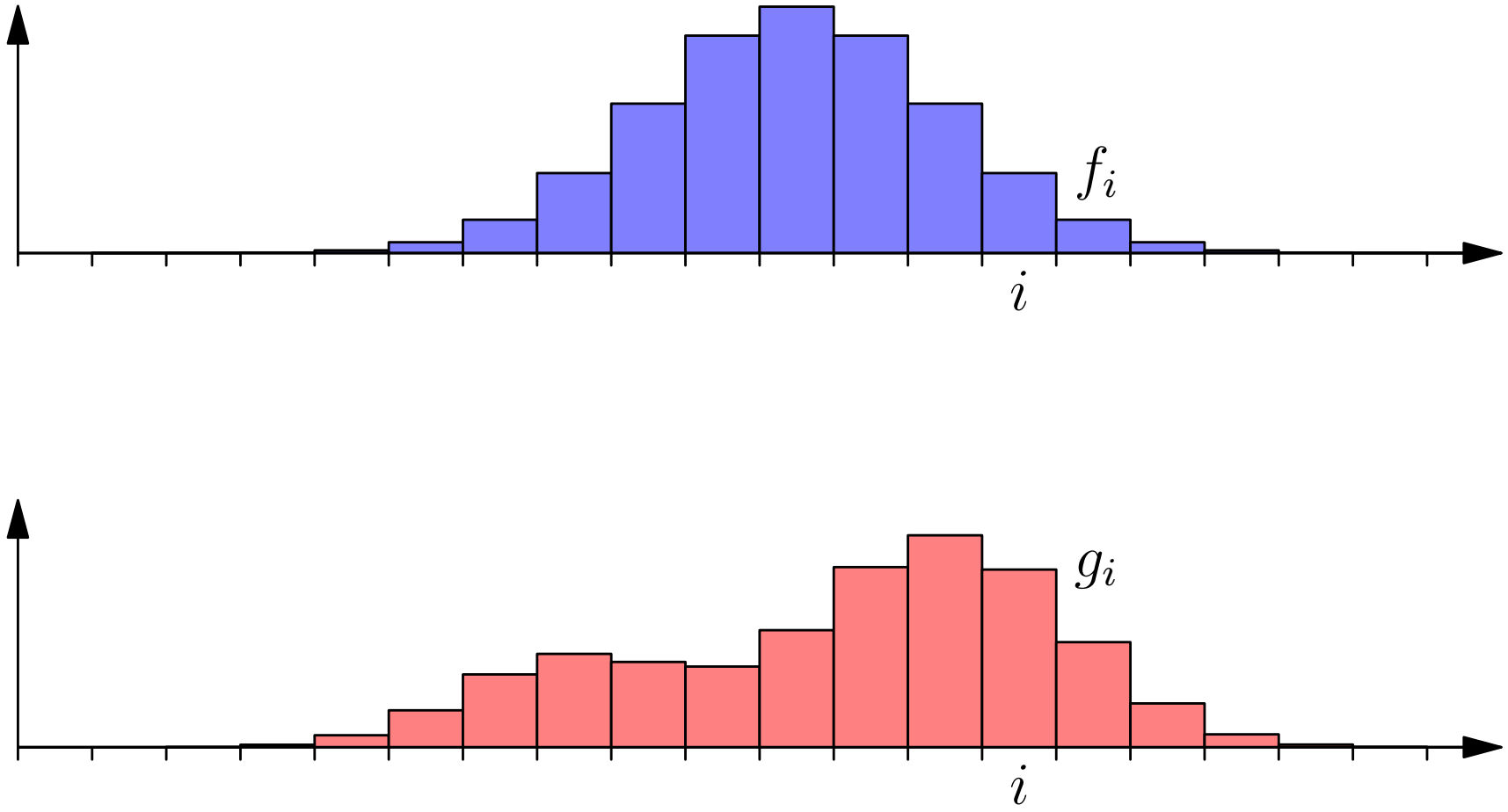
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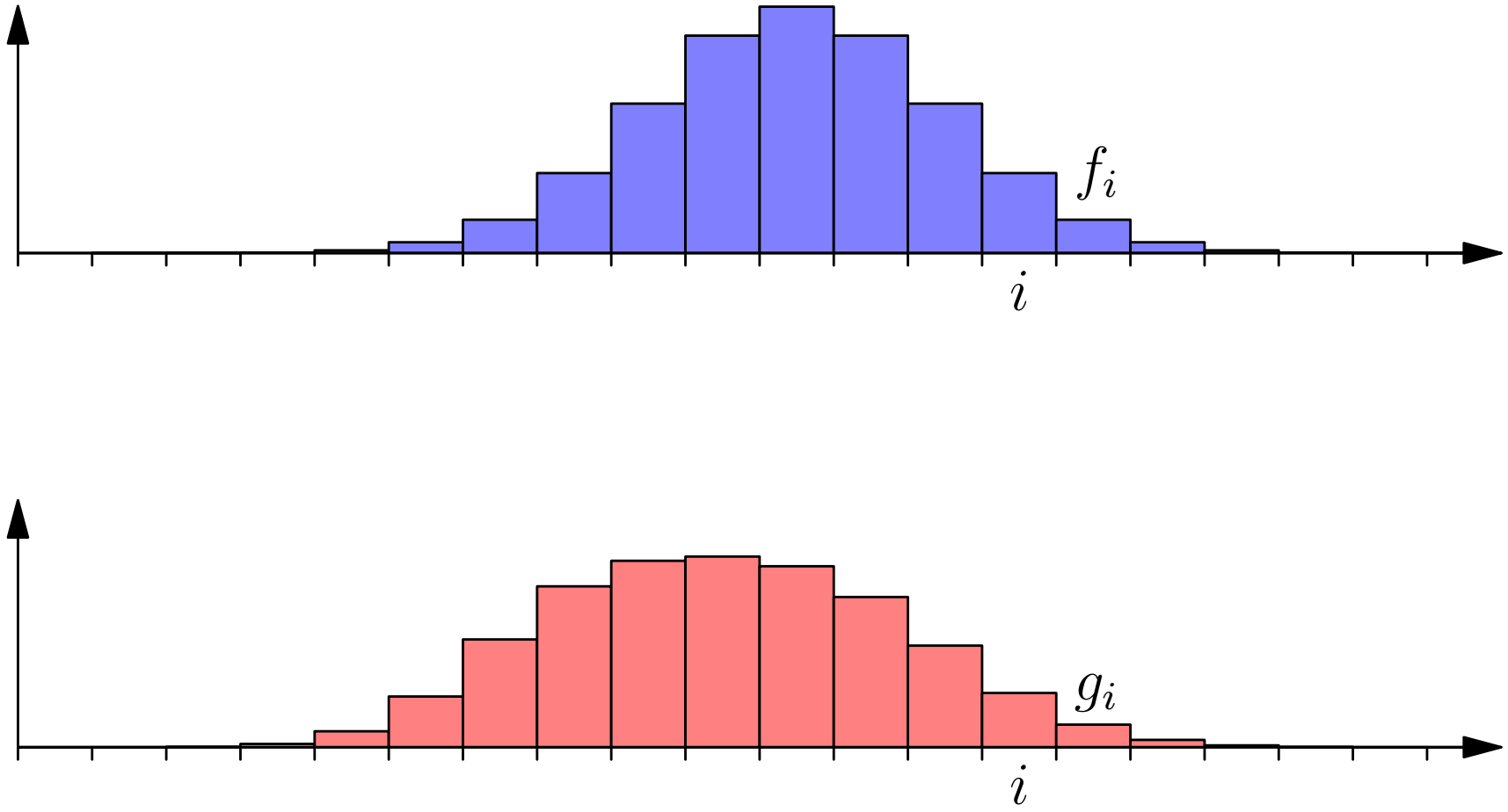
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- We will meet KL-divergences later on

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